Criterion A

Description of Scenario.

My client is Dr. Manuel Bazaldua, a general surgeon who mainly works by giving private consultations to his patients in two consulting rooms in different hospitals. As my client described during the interview, currently, the process that my client follows when scheduling an appointment with a patient is through his assistant. Depending on whether the patient has been in an appointment with him before or it is their first time, his assistant takes the patient's contact information in her agenda, their medical history by hand on a template, and schedules the consultation in a separate app. The greatest difficulty my client passes through is that all the information about a patient and their consultation ends up in different places, which causes a feeling of lack of organization in the consultation process. Dr. Bazaldua is not able to see all the essential details about consultations in the same place and has to look for them in all of these different reports, which makes this process tedious and time-consuming. My client also has a hard time predicting how many appointments he schedules depending on the time of the month, which makes it difficult for him to plan vacations or days off while keeping the amount of losses as low as possible.

Rationale for the proposed solution

Firstly, for this project, the programming language I decided to use is Java. This is mainly because this is an object-oriented language, which I find is the language that best fits with how I wanted my application to work and my objectives. Java's OOP characteristics will enable me to associate certain information with a specific object, in the case of my project, a patient, and will make the development of other features of my application more organized and efficient. Java also facilitates the creation of separated files with crucial information for the user in an organized and legible way. In the case of my project, the list of appointments will be displayed in my app, but the user will also be able to access the list and save this information in an auto-completed CSV file on their computer. I decided to use JavaFX and Scene Builder because these tools will allow me to develop a GUI application easily. These tools also include a complete amount of features that will contribute to my application being easier to comprehend and manage by my user, by either using different types of input options to specify the information required or including more visually appealing ways to portray information, for example using a Table View or graphs. Finally, I specifically decided to use JavaFx8 since I found that many useful features are included in this version, one of the most important for my project is the "Date Picker". The "Date Picker" will be used several times in crucial parts of the project.

For example, when generating automatically the age of my patient and when wanting to set an appointment on a specific date.

Success Criteria.

- The app includes a security system, in which the client has to log in with their username and password.
- The client can add new patients with their information
- The client can edit or delete patients registered.
- The client can search easily by name for a specific patient.
- The client can schedule a consultation with registered patients.
- The program will automatically generate a txt file with the patient information the client requires.
- All data input follows the required type of information established by my client.
- The client will be able to see graphical trends in the frequency of their appointments.
- The clients will be able to see graphically which doctor refers more clients.

Word Count: 497